

Instagram Satisfaction: A Sentiment & Rating Analysis

Project Report

Team 6B, Dream Team:

Naomi Hossain

Xinyi He

Luisa Garate

Weinou Jin

Tysean Haigood

Table of Contents

1. Introduction	3
1.1 Background	3
1.2 Problem Statement	3
1.3 Research Questions	3
2. Related Work	4
3. Data	4
3.1 Data Source	4
3.2 Variables Description	5
3.3 Data Cleaning & Processing	6
4. Exploratory Analysis	6
5. Analysis of Findings	8
6. Conclusion	10
References	12

1. Introduction

1.1 Background

In today's digital landscape, user-generated feedback plays a critical role in shaping mobile app development and customer engagement strategies. As one of the world's most widely used social media platforms, Instagram receives thousands of App Store reviews daily, reflecting real-time user experiences, frustrations, and expectations. Analyzing these reviews through sentiment analysis provides valuable insights into how users perceive recent updates, interface changes, and feature performance.

As mobile applications continue to evolve, user feedback has become a valuable resource for understanding product performance and shaping development decisions. App Store reviews, in particular, offer authentic, unsolicited insights into how users perceive an app's functionality, design, stability, and overall value. Instagram is one of the most prominent social media platforms globally. Its frequent updates, features, interface, and performance to meet the expectations of millions of daily users. These continual changes generate a steady stream of reviews that reflect user reactions, including praise, complaints, and suggestions.

1.2 Problem Statement

Understanding how user sentiment varies across geographic regions is essential for evaluating the global performance of digital platforms. As Instagram continues to expand its international user base, the App Store reviews left by users provide valuable insight into how experiences, expectations, and reactions differ across countries. This project investigates how sentiment toward Instagram's mobile application has evolved across different locations, analyzing patterns in user feedback to uncover regional differences, cultural factors, and potential issue hotspots. By examining sentiment trends by country, this analysis offers a deeper understanding of global user experiences and provides actionable insights that can support Instagram's product development, localization strategies, and customer engagement efforts.

1.3 Research Questions

This project investigates the following questions:

- How do star ratings vary across different countries?
- How does text sentiment differ from numerical ratings?
- What themes appear in positive vs. negative reviews?
- Which regions show higher or lower user satisfaction?

2. Related Work

We were inspired to pursue this project by a desire to better understand Instagram's global success and the factors that have made it a transformative platform in people's daily lives. As a leading social media application, Instagram plays a significant role in communication, content creation, and digital culture, but its user experience may not be consistent across regions. To explore whether the app performs similarly in different countries and how user perceptions vary internationally, we chose to focus our analysis on App Store reviews. These reviews offer authentic, user-generated insights that allow us to examine sentiment across locations and identify meaningful differences in satisfaction, expectations, and overall app performance.

3. Data

3.1 Data Source

The raw data for this project was obtained by us from a publicly available dataset on the Kaggle platform (Kanchana1990, 2023). This dataset was collected by the author from real reviews on the Apple App Store's review pages and has been organized by the author into a structured CSV format. Since the original publisher has already completed the data collection and anonymization process, we started working directly by downloading this dataset and did not involve the web scraping step.

3.2 Variables Description

The dataset contains 500 user reviews published within seven days, covering multiple countries and app store regions. It includes **11 variables**, such as:

appId	a unique id from apple
country	the country/region associated with the reviewer's App Store
date	the date the user posted the review
id	a unique id
score	the star rating given by the user (1-5)
text	the full written content of the review
title	the short headline/title the user gives to their review
url	url of the review
userName	anonymized username of the reviewer as it appears in the App Store
urserURL	url of the user
version	the version of the app that user is reviewing

These fields not only provide sufficient basis for sentiment analysis but also facilitate processing in programming projects.

3.3 Data Cleaning & Processing

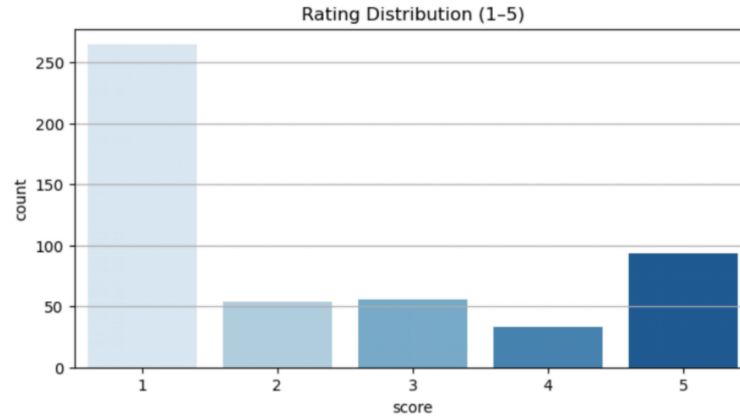
Before the formal analysis began, we first carried out necessary preprocessing on the data to ensure that all data met the requirements of usability and standardization. The first step was to

extract the key variables relevant to the research question from the dataset. These variables mainly included the comment date, the user's region, the given rating, and the comment text. Then, the date column was converted to a standard date format, the star ratings were unified into numerical types, and the records that could not be parsed were deleted to enable the program to normally input standardized information. Since some comments only contained titles or main texts, we merged these two fields into complete comment texts to ensure that each record had a uniform format and complete content. To further support the analysis, we also constructed time-based derived variables, such as the comment year and the combination of year and month.

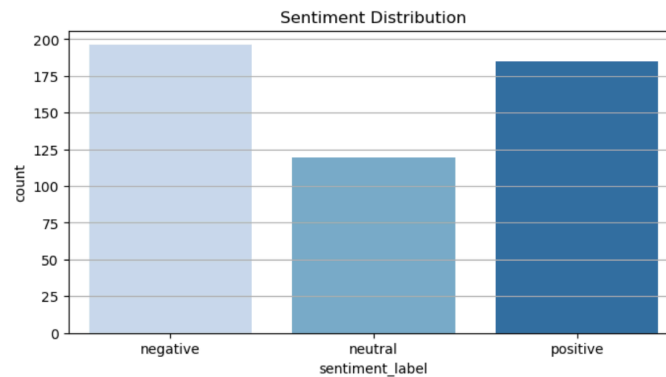
In the text processing stage, we systematically cleaned all the comment texts. The specific operations included converting all texts to lowercase, thoroughly removing URL links, non-alphabetic characters and extra spaces. These treatments significantly improved the format consistency of the text data. The cleaned texts provided a reliable foundation for the subsequent input stage, effectively reducing noise interference, and thus enabling high-accuracy sentiment analysis and word frequency statistics results. After the above steps, the dataset has reached a clear and usable state, which not only supports statistical exploration but also effectively supports text-level insights.

4. Exploratory Analysis

In this project, exploratory analysis aims to understand users' rating behavior and the language they use in text comments. We start with basic statistical summaries, observing the star distribution of all comments and those from different countries. Specific analyses include frequency statistics, average score calculation, and comparisons of rating patterns in different regions. These initial statistics help determine whether there are systematic tendencies of overly high or low ratings in certain regions.



To gain deeper insights from the data, we incorporated a text-based approach to sentiment analysis. Since numerical ratings alone can overlook subtle shades of user sentiment, analyzing the text of reviews helps reveal those nuances. For this task, we selected the VADER sentiment model (Hutto & Gilbert, 2014), this is a rule-based tool particularly well-suited for short, informal texts like social media content. This model generates a sentiment score for each comment and classifies it as "positive", "neutral", or "negative". With VADER, we compared the text sentiment with the star ratings to identify comments where the two were inconsistent, such as those with positive words but low ratings, or negative tone but high ratings.



We also constructed a cross-table and combined it with a heatmap for visualization, in order to explore the correlation between the ratings and the sentiment labels. This comparison can reveal whether the content of user comments is consistent with the given ratings. Additionally, we calculated national-level statistics, such as the average sentiment score and average star rating, to reveal the regional distribution pattern of user satisfaction. These aggregated data go beyond the differences of individual comments and help us understand the overall perception of the application by users in different regions.

In terms of text analysis, we also extracted frequent words from both positive and negative comments. After completing the text cleaning process, including removing punctuation, URLs, and non-alphabetic characters, we generated word clouds for each sentiment category. Word clouds were generated using the Python WordCloud library (Mueller, 2020). Although word clouds do not possess the rigor of statistical methods, as an intuitive exploration tool, they can effectively reveal common discussion topics, frequently occurring issues, as well as the functional points that users repeatedly praise or criticize.

Overall, this analysis integrates the quantitative data from user feedback with the qualitative content to form a complete analytical framework. By combining descriptive statistics, sentiment scoring, regional comparisons, and text exploration, we have constructed a multi-dimensional analysis framework. This framework provides a more comprehensive perspective for understanding how users experience the Instagram application and the differences in user evaluations across different regions. This framework offers a more comprehensive perspective for understanding the differences in user experience and evaluation of Instagram across various regions.

5. Analysis of Findings

Based on the preliminary exploratory analysis, we comprehensively employed methods such as statistical summaries, sentiment scoring, and cross-national comparisons to conduct an in-depth analysis of the regional differences in user experience. Firstly, we observed the distribution of star ratings, which helped us understand the general outline of users' surface satisfaction. Although the ratings cover the full range from 1 to 5, the scores alone cannot explain the specific reasons behind users' evaluations, nor can they determine whether the same rating has a similar meaning in different countries.

To overcome this limitation, we incorporated the sentiment scoring results of the VADER model. By calculating the sentiment score for each comment and classifying them, we compared and analyzed the emotional tone of the text with the numerical scores. When cross-analyzing the scores with the sentiment labels, some interesting phenomena emerged: a considerable number of

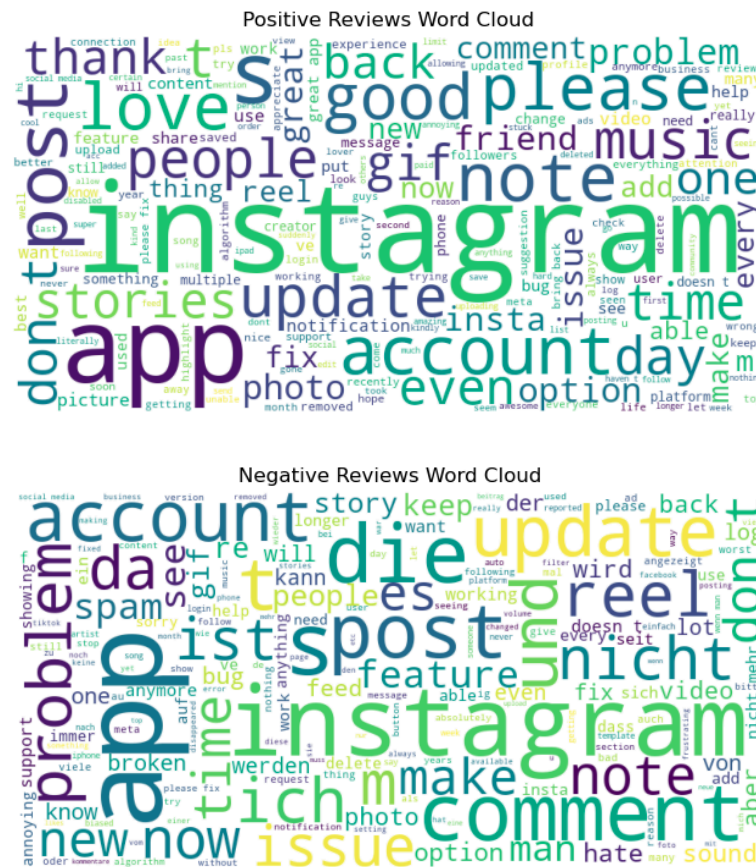
low-scoring comments actually used neutral or even slightly positive wording, while some high-scoring comments contained expressions of dissatisfaction. This inconsistency indicates that relying solely on the scores is not sufficient to fully reflect the user's feelings; text sentiment analysis provides an additional interpretive dimension.

After obtaining the emotional scores, we further analyzed the differences in performance of various countries in both the rating and emotional dimensions. By calculating the average star ratings and sentiment scores for each region, we discovered that these differences follow a clear pattern. Some countries consistently show a more positive evaluation tendency, while the satisfaction levels in other regions are relatively lower. Taking Germany as an example, its average rating and sentiment score are the lowest, indicating that German users encountered more problems during their usage, or were more inclined to express dissatisfaction in their comments. In contrast, users in India and the United States had the highest satisfaction, and this result was confirmed in both the rating and text emotional dimensions. Countries like Canada and the United Kingdom showed a middle state where the digital rating and emotional score were not completely consistent, reflecting a more complex user attitude.

***	country	avg_score	avg_sentiment	review_count
1	DE	1.81	-0.237684	100
2	GB	2.27	-0.008721	100
0	CA	2.18	0.000460	100
3	IN	2.60	0.108436	100
4	US	2.51	0.111787	100

In order to ultimately clarify the specific concerns of the users, we turned to the word frequency analysis of the comment texts. After classifying the texts by sentiment and extracting the frequent words for visualization, the core themes of positive and negative evaluations became clearly apparent. For instance, words such as "application", "dynamic stories", and "music" frequently appeared in positive comments, indicating the users' high approval of these social and creative functions. Negative comments, on the other hand, focused on words like "account", "problems", "failures", "updates", and "short videos", pointing to technical issues such as login difficulties, program errors, or functional abnormalities. Although this is not a complete theme

modeling, it does provide an intuitive reference for us to understand the driving factors behind users' satisfaction and dissatisfaction.



Overall, this method that combines statistical analysis with text interpretation is more capable of comprehensively presenting user opinions than a single analytical approach. The discovered patterns in this stage have also laid a solid foundation for the final conclusions and recommendations of the project.

6. Conclusion

When analyzing the feedback from Instagram users in various countries, we found that the star ratings given by the users often did not match the emotions expressed in their comments. This phenomenon is quite thought-provoking: many low-star ratings appear neutral or even positive in their words, while some high-star evaluations actually reveal complaints or dissatisfaction.

From the perspective of geographical distribution, the satisfaction levels of users in different countries show a distinct pattern of differences. The ratings and sentiment scores from German users were the lowest, indicating the most negative state. The comments from users in India and the United States are much more positive. This usually indicates that users in these regions encounter fewer obstacles during the application usage process. The UK and Canada were in the middle range, with their ratings and sentiment expressions not being completely synchronized. The users from these different regions, as we studied them, produced a strong and interesting contrast. This phenomenon demonstrates that users' attitudes may be influenced by fluctuations and may be accompanied by complex situations.⁶

By conducting a further analysis of the comment content, we identified the key factors influencing users' emotions. Positive evaluations usually revolve around functions such as "social interaction" and "creative expression"; while negative comments frequently mention technical issues like "account anomalies", "update failures", and "function malfunctions". These frequently-occurring words help us clearly see in which aspects the product is recognized and in which areas it frequently causes users' complaints.

Overall, this study has demonstrated the effectiveness of integrating statistical methods with natural language processing. This hybrid analysis approach not only reveals the limitations of the scoring system but also helps us extract more actionable insights from user text. Ultimately, we are able to more accurately determine which issues have the greatest impact on user experience, thereby providing a clear direction for product optimization - allocating resources to the aspects that users are most concerned about and where feedback is most concentrated.

7. Project Slides

[Slides created by Team 6B, Dream Team](#)

References

Hutto, C. J., & Gilbert, E. (2014). *VADER: A parsimonious rule-based model for sentiment analysis of social media text*. In *Proceedings of the International AAAI Conference on Web and Social Media* (pp. 216–225).

<https://ojs.aaai.org/index.php/ICWSM/article/view/14550>

Kanchana1990. (2023). *Instagram ratings and reviews (App Store) [Data set]*. Kaggle.

<https://www.kaggle.com/datasets/kanchana1990/instagram-ratings-and-reviews-appstore>

McKinney, W. (2010). *Data structures for statistical computing in Python*. *Proceedings of the Python in Science Conference*, 1–12.

<https://conference.scipy.org/proceedings/scipy2010/pdfs/mckinney.pdf>

Mueller, A. (2020). *Word cloud generation in Python*. Python Software Foundation.

https://github.com/amueller/word_cloud